

Runchu Tian

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EDUCATION

University of Illinois Urbana-Champaign

Master of Science in Computer Science

GPA: 4.00/4.00

Sep. 2024 - Present

Tsinghua University

Bachelor of Science in Mathematics and Physics

GPA: 3.80/4.00

Sep. 2020 - Jun. 2024

RESEARCH PUBLICATIONS

Tsinghua University, Natural Language Processing Lab

Mar. 2022 - Jul. 2024

Adviser: Zhiyuan Liu, Associate Professor, Department of Computer Science and Technology

Title: *DebugBench: Evaluating debugging capability of large language models*

ACL 2024

Authors: Runchu Tian*, Yining Ye*, Yujia Qin, Xin Cong, Yankai Lin, Yinxu Pan, Yesai Wu, Haotian Hui, Weichuan Liu, Zhiyuan Liu, Maosong Sun

Overview: Introduced a benchmark of 4,253 instances across multiple programming languages and bug types to systematically evaluate and analyze the debugging capabilities of large language models.

Sept. 2023 - Feb. 2024

Title: *Distance between Relevant Information Pieces Causes Bias in Long-Context LLMs*

ACL 2025

Authors: Runchu Tian*, Yanghao Li*, Yuepeng Fu, Siyang Deng, Qinyu Luo, Cheng Qian, Shuo Wang, Xin Cong, Zhong Zhang, Yesai Wu, Yankai Lin...

Overview: Introduced a benchmark for evaluating positional bias in large language models when handling multiple pieces of relevant information, revealing the significant bias based on the spacing of relevant information.

May 2024 - Aug. 2024

Title: *Exploring Format Consistency for Instruction Tuning*

TMLR

Authors: Shihao Liang*, Runchu Tian*, Kunlun Zhu*, Yujia Qin, Huadong Wang, Xin Cong, Zhiyuan Liu, Xiaojiang Liu, Maosong Sun

Overview: Introduced a framework that automatically transfers instruction formats across datasets to ensure format consistency in instruction tuning, showing that consistent formats improve model performance.

Jan. 2023 - June 2023

Title: *ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs*

ICLR 2024 spotlight

Authors: Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, Maosong Sun

Overview: Proposed a comprehensive framework for enabling large language models to use external APIs through automated dataset construction (ToolBench), a reasoning-enhanced training algorithm, an evaluation suite (ToolEval) and a language model (ToolLLaMA), which achieves ChatGPT-level tool-use performance.

Mar. 2023 - Jun. 2023

Title: *Tool Learning with Foundation Models*

ACM Computing Surveys, 2024

Authors: Yujia Qin, Shengding Hu, Yankai Lin, Weize Chen, Ning Ding, Ganqu Cui, Zheni Zeng, Yufei Huang, Chaojun Xiao, Chi Han, Yi Ren Fung, Yusheng Su, Huadong Wang, Cheng Qian, Runchu Tian, Kunlun Zhu, Shihao Liang, Xingyu Shen ...

Overview: Provided a comprehensive survey and framework for tool learning with foundation models.

Jan. 2023 - May 2023

University of Illinois Urbana-Champaign, Data Mining Group

Sep. 2024 - Present

Adviser: Jiawei Han, Michael Aiken Chair Professor, Siebel School of Computing and Data Science

Title: *LLM-Based Compact Reranking with Document Features for Scientific Retrieval*

Under Review (KDD 2026)

Authors: **Runchu Tian**, Xueqiang Xu, Bowen Jin, SeongKu Kang, Jiawei Han
Overview: Proposed a compact form of document representation in LLM-based listwise reranking. May 2025 - Aug. 2025

Title: *Beyond True or False: Retrieval-Augmented Hierarchical Analysis of Nuanced Claims* ACL 2025
Authors: Priyanka Kargupta*, **Runchu Tian***, Jiawei Han
Overview: Developed ClaimSpect, a retrieval-augmented framework that hierarchically deconstructs nuanced claims into structured aspect trees enriched with corpus-specific evidence and perspectives for in-depth analysis. Oct. 2024 - Feb. 2025

Title: *PairSem: LLM-Guided Pairwise Semantic Matching for Scientific Document Retrieval* WWW 2026
Authors: Wonbin Kweon, **Runchu Tian**, SeongKu Kang, Pengcheng Jiang, Zhiyong Lu, Jiawei Han, Hwanjo Yu
Overview: Developed PairSem, an unsupervised LLM-guided framework that models scientific concepts as entity–aspect pairs to enable fine-grained, retriever-agnostic scientific document retrieval. May 2025 - Aug. 2025

Title: *Retrieval and Structuring Augmented Generation with Large Language Models* KDD 2025 Conference Tutorials
Authors: Pengcheng Jiang, Siru Ouyang, Yizhu Jiao, Ming Zhong, **Runchu Tian**, Jiawei Han
Overview: Reviewed the critical role of structured knowledge in RAG. Nov. 2024 - Jun. 2025

Title: *Deepretrieval: Hacking real search engines and retrievers with large language models via reinforcement learning* COLM 2025
Authors: Pengcheng Jiang, Jiacheng Lin, Lang Cao, **Runchu Tian**, SeongKu Kang, Zifeng Wang, Jimeng Sun, Jiawei Han
Overview: Explored GRPO-based methods for optimizing LLM query rewriting for search engines. Dec. 2024 - Jul. 2025

Title: *Topic Coverage-based Demonstration Retrieval for In-Context Learning* EMNLP 2025
Authors: Wonbin Kweon, SeongKu Kang, **Runchu Tian**, Pengcheng Jiang, Jiawei Han, Hwanjo Yu
Overview: Proposed TopicK, a topic coverage-based retrieval framework that selects demonstrations by modeling the fine-grained topical knowledge needs for effective in-context learning. Feb. 2025 - May 2025

University of California San Diego, Shang Data Lab (as visiting student) Jun. 2025 - Aug. 2025
Adviser: Jingbo Shang, Associate Professor, Department of Computer Science and Halicioğlu Data Science Institute

Title: *Finish First, Perfect Later: Test-Time Token-Level Cross-Validation for Diffusion Large Language Models* Under Review (ICML 2026)
Authors: **Runchu Tian***, Junxia Cui*, Xueqiang Xu, Feng Yao, Jingbo Shang
Overview: Introduced a test-time decoding strategy that enables token-level cross-validation and iterative refinement in diffusion LLMs, significantly improving generation quality across diverse language tasks. Jul. 2025 - Oct. 2025 (continued remotely)

* denotes equal contribution

PROFESSIONAL EXPERIENCE

University of Illinois Urbana-Champaign, Siebel School of Computing and Data Science

Teaching Assistant, CS 173 - Discrete Structures Sept. 2024 – Dec. 2024

Research Assistant, on *Scientific Claim Verification* advised by Prof. Jiawei Han Jan. 2025 – May 2025

Research Assistant, on *Geospatial Understanding Information System* advised by Prof. Jiawei Han May 2025 – Present

HONORS AND AWARDS

Tsinghua University Academic Excellence Scholarship	2021
Tsinghua University Academic Excellence Scholarship	2022